TARGET SPEECH EXTRACTION ON EDGE DEVICE

A project report submitted in partial fulfilment
of the requirements for the degree of

Bachelor of Technology
in
Electronics and Computer Engineering
by

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Abstract

Hearing aids, earbuds, conferencing devices and voice assistants must often pick out one voice from several people talking at once. Target speech extraction (TSE) solves this by using a short enrollment recording of the wanted speaker to extract only that speaker's voice from a mixture. To be useful in such products, TSE must run on the device itself, for privacy and low latency, and it must keep working when more than two people talk. Most published TSE systems are evaluated only on two-speaker mixtures and on server hardware.

This project designs, trains and deploys a time-domain TSE network for edge devices. The model combines a U-shaped multi-resolution convolutional separator with speaker conditioning by feature-wise linear modulation, speaker-gated feature splitting and gated multi-stage aggregation. One model is trained on dynamically generated mixtures of two, three and four speakers drawn from LibriSpeech, and is evaluated on Libri2Mix, Libri3Mix and a four-speaker extension. On these test sets it improves the scale-invariant signal-to-distortion ratio by 9.49, 8.13 and 7.00 dB respectively, while a model of the same design trained only on two-speaker data reaches 4.84 and 3.13 dB on three and four speakers. For deployment, the network is split into a once-per-speaker embedder and a streaming extractor, exported to ONNX and quantised to 8-bit integers, which shrinks the model from 51 MB to 14 MB within 0.13 dB of the full-precision model. A sliding-window streaming scheme provides real-time operation. In an offline Android application on a commercial smartphone, the model processes speech 13 times faster than real time with the same quality as on the PC, and streams in real time with one second of latency; the compact variant streams with 0.4 s of latency even on a single CPU core. Multi-head and differential attention variants of the bottleneck were also studied. The results show that accurate multi-speaker target speech extraction is feasible within the memory and compute budget of edge hardware.

Keywords: target speech extraction, speaker conditioning, multi-speaker mixtures, edge deployment, INT8 quantisation, ONNX, streaming inference.

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We thank our parents, family, and friends for bearing with us throughout the course of our project and for the opportunity they provided us in undergoing this course in such a prestigious institution.

Codes and Standards

Project Title: **Target Speech Extraction on Edge Device**

1. Speech Quality and Telephony Standards

ITU-T P.862 - Perceptual Evaluation of Speech Quality (PESQ)
Objective speech-quality measure used to score the extracted speech (narrow-band mode, 8 kHz).

ITU-T P.862.1 - Mapping function for PESQ to MOS-LQO
Maps raw PESQ scores to the mean-opinion-score scale reported in this work.

ITU-T G.711 - Pulse Code Modulation of voice frequencies
Defines the 8 kHz narrow-band telephony bandwidth used for all audio in this project.

2. Data Formats and Model Interchange

ONNX (Open Neural Network Exchange), operator set 17
Open standard used to export the trained network to a hardware-independent edge runtime.

IEEE 754-2019 - Floating-point arithmetic
Number format of the FP32/FP16 models; INT8 quantisation is defined relative to it.

RIFF/WAVE PCM audio format
Container format of all speech recordings, mixtures and model outputs.

3. Software Quality and Testing

ISO/IEC 25010 - Systems and software quality model
Guides the functional suitability, performance efficiency and maintainability goals of the software.

ISO/IEC/IEEE 29119 - Software testing
The pipeline is covered by automated unit and integration tests (shapes, metrics, checkpoints, streaming).

4. Data Protection and Security

Digital Personal Data Protection (DPDP) Act, 2023 (India)
Voice recordings are personal data; on-device processing keeps speech and enrollment data on the device.

ISO/IEC 27001 - Information security management
Applies to storage of the speaker enrollment embedding on the device.

5. Hardware Safety (edge device)

IEC 62368-1 - Audio/video, information and communication technology equipment safety
Safety requirements for the battery-powered or mains-powered audio device hosting the model.

Android Compatibility Definition / Google Play app policies
The application uses only public Android APIs and requests only the microphone permission.

Mapping of Sustainable Development Goals (SDGs)

Project Title: **Target Speech Extraction on Edge Device**

Mapped SDG	Justification	Example Impact
SDG 3 - Good Health and Well-being	Extracting the voice a listener wants to hear is the core function of a hearing aid in a noisy, multi-talker place.	Better speech understanding for people with hearing loss in restaurants, classrooms and family gatherings.
SDG 10 - Reduced Inequalities	On-device processing works offline on low-cost hardware, without paid cloud services.	Assistive listening becomes affordable and available where network access is poor.
SDG 4 - Quality Education	Isolating the teacher's voice from classroom babble improves lecture audio and captions.	Clearer recorded and live lectures for students, including those with hearing or attention difficulties.
SDG 9 - Industry, Innovation and Infrastructure	Compact, quantised neural speech processing that runs on edge processors.	A reusable, open pipeline (training, export, quantisation, streaming) for embedded speech products.

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List of Acronyms

Acronym	Full Form
TSE	Target Speech (Speaker) Extraction
BSS	Blind Source Separation
SI-SDR	Scale-Invariant Signal-to-Distortion Ratio
SI-SDRi	SI-SDR improvement over the mixture
SDR / SIR / SAR	Signal-to-Distortion / -Interference / -Artifact Ratio
SNR	Signal-to-Noise Ratio
PESQ	Perceptual Evaluation of Speech Quality
STOI	Short-Time Objective Intelligibility
NSR	Negative-SI-SDRi Rate (wrong speaker or failed extraction)
FiLM	Feature-wise Linear Modulation
GES	Gated Embedding Split
MGFA	Multi-stage Gated Feature Aggregation
SE	Squeeze-and-Excitation (channel attention)
MHSA	Multi-Head Self-Attention
DIFF	Differential attention
gLN	global Layer Normalisation
DS-conv	Depthwise-Separable convolution
PReLU	Parametric Rectified Linear Unit
CE	Cross-Entropy
ECAPA-TDNN	Emphasized Channel Attention, Propagation and Aggregation TDNN
ONNX	Open Neural Network Exchange
ORT	ONNX Runtime
FP32 / FP16 / INT8	32-bit float / 16-bit float / 8-bit integer number formats
RTF	Real-Time Factor (processing time / audio duration)
LR	Learning Rate
CPU / GPU	Central / Graphics Processing Unit
DNN	Deep Neural Network
ASR	Automatic Speech Recognition
MOS	Mean Opinion Score
STFT	Short-Time Fourier Transform

CHAPTER 1
Introduction

1.1 Background

In everyday environments speech rarely arrives alone. A listener in a restaurant, a meeting room or a family gathering hears several people talking at the same time, together with the target talker. Human listeners can attend to one voice and ignore the others, an ability known as the *cocktail-party effect* [1]. Machines and hearing-impaired listeners find this much harder: hearing aids amplify all talkers equally, and speech recognisers and voice assistants make many more errors when another person speaks over the user.

Two families of deep-learning methods address overlapping speech. *Blind source separation* (BSS) splits a mixture into all of its speakers [7], [9], [10], but it must know how many speakers there are, it returns the sources in an arbitrary order, and its cost grows with the number of outputs. *Target speech extraction* (TSE) instead uses a short enrollment recording of the wanted speaker and returns only that speaker's voice [2]-[6]. TSE answers the question a device actually needs answered ("what is my user saying?") with one output, whatever the number of interfering talkers.

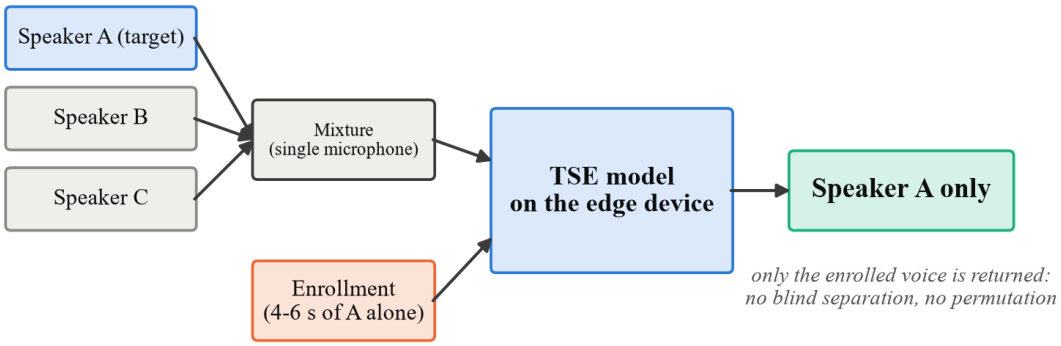


Figure 1.1: Target speech extraction: a short enrollment of speaker A selects which voice is extracted from a single-microphone mixture of several talkers

1.2 Motivation for On-Device Processing

The products that need TSE most are small, battery-powered devices: hearing aids, earbuds, smart speakers, conference-room units and mobile phones. Running the model on the device itself, at the edge, rather than in the cloud has four advantages:

- **Privacy:** speech and the user's enrollment never leave the device, which matters for data-protection law and user trust.

- **Latency:** a hearing aid or a call must respond within a few hundred milliseconds; a network round-trip alone can exceed this.
- **Availability:** the device keeps working without network coverage and without running costs.
- **Scalability:** processing cost is spread over the devices instead of a central server.

Edge hardware, however, has limited memory, compute and power. A TSE model for the edge must therefore be small, cheap to run per second of audio, robust to reduced numerical precision, and able to process a continuous audio stream.

1.3 Problem Statement

Given a single-channel mixture $y = s_t + \Sigma\, s_i$ of a target speaker and one to three interfering speakers, and a separate enrollment utterance of the target speaker, estimate the target speech s_t with high fidelity, using a model that fits the memory and compute budget of an edge device and can run in real time on streaming audio. The number of interfering speakers is not known in advance.

1.4 Objectives

1. Build a complete, reproducible TSE pipeline: data indexing, enrollment selection, training, evaluation and inference, with no information leaking between data splits.
2. Design a compact time-domain TSE network with strong speaker conditioning.
3. Train one model that handles two, three and four simultaneous speakers, and evaluate it separately on each case.
4. Evaluate quality with signal metrics (SI-SDR, SNR, SDR, SIR, SAR) and perceptual metrics (PESQ, STOI).
5. Prepare the model for edge deployment: graph splitting, ONNX export, INT8 quantisation and streaming inference, and measure model size, compute, latency and quality after each step.
6. Study whether multi-head or differential attention at the separator bottleneck improves the model.

1.5 Contributions

- A U-shaped convolutional TSE network (U-eConv-TSE) with FiLM speaker conditioning, gated embedding splitting and multi-stage gated aggregation; the deployed model has 13.4 M parameters.
- Dynamic 2-4 speaker training, with which one model reaches 9.49 / 8.13 / 7.00 dB SI-SDRi on 2 / 3 / 4-speaker test sets, against 9.61 / 4.84 / 3.13 dB for the same design trained on two speakers only.
- An edge deployment path (embedder/extractor split, ONNX, INT8, sliding-window streaming) with measured size, latency and accuracy at every step.
- An evaluation suite with eight metrics, a numpy implementation of BSS-eval verified against the reference implementation, and a breakdown of the remaining errors.
- A controlled study of multi-head and differential attention in the separator bottleneck.

1.6 Organisation of the Report

Chapter 2 reviews prior work on speech separation, target speech extraction, conditioning and attention mechanisms, and efficient on-device speech processing, and lists the gaps this project addresses. Chapter 3 describes the model, the data pipeline, the training procedure, the evaluation metrics and the edge deployment pipeline. Chapter 4 presents and discusses the results: quality for two to four speakers, error analysis, the attention study and the edge deployment measurements. Chapter 5 concludes the report and outlines future work.

CHAPTER 2
Literature Survey

2.1 Review of Prior Work

2.1.1 Blind Speech Separation

Deep clustering [9] learned time-frequency embeddings that are clustered into speakers, and permutation-invariant training (PIT) [10] solved the label-ordering problem of training a network with several outputs. Conv-TasNet [7] replaced the short-time Fourier transform by a learned convolutional encoder and decoder and estimated masks with a temporal convolutional network, surpassing ideal time-frequency masks on WSJ0-2mix. Dual-path RNN [12] and SepFormer [11] model long sequences with recurrent and Transformer [15] blocks. SuDoRM-RF [8] showed that U-shaped stacks of depthwise-separable convolutions that successively down-sample and up-sample the signal reach competitive quality at a fraction of the compute, which makes this structure attractive for edge devices. All BSS models need the number of speakers and return outputs in an arbitrary order.

2.1.2 Target Speech Extraction

SpeakerBeam [2] introduced speaker-aware networks in which an auxiliary network turns an enrollment utterance into an adaptation vector that modulates the extraction network. Time-domain SpeakerBeam [3] moved this idea to a Conv-TasNet-style network and added a speaker-identification loss to sharpen speaker discrimination. VoiceFilter [4] conditions a spectrogram-masking network on a d-vector, and VoiceFilter-Lite [5] reduced it to a streaming, quantised model for on-device speech recognition, which is the closest prior work to this project's deployment goal. SpEx+ [6] extracts the target with a fully time-domain multi-scale network and shares the encoder between the mixture and the enrollment.

2.1.3 Speaker Conditioning and Attention

Feature-wise linear modulation (FiLM) [13] conditions a network by predicting a per-channel scale and shift from a conditioning vector, a cheap and effective way to inject the speaker identity. Squeeze-and-excitation (SE) [14] re-weights channels using a global summary of the input. Self-attention [15] relates every time step to every other one; the Differential Transformer [16] computes attention as the difference of two softmax maps, which cancels attention mass that both maps share and has been shown to reduce attention noise in language models.

2.1.4 Speaker Embeddings

x-vectors [17] and ECAPA-TDNN [18] are speaker-verification networks whose utterance-level embeddings are widely used as speaker cues. They are trained on thousands of speakers, but they are not trained to guide an extraction network; this project compares them with a jointly trained speaker encoder.

2.1.5 Datasets and Evaluation

LibriSpeech [19] provides read English speech from over two thousand speakers. LibriMix [20] builds Libri2Mix and Libri3Mix mixtures from it and is the standard open benchmark for separation and extraction. Quality is measured with the scale-invariant SDR [21], the BSS-eval decomposition into distortion, interference and artifacts [22], PESQ [23] and STOI [24].

2.1.6 Efficient On-Device Inference

Integer quantisation [25] stores weights and computes with 8-bit integers, reducing model size four-fold and speeding up inference on processors with integer vector units. ONNX and ONNX Runtime [26] provide a hardware-independent model format and an optimised runtime for CPUs, mobile and embedded devices. VoiceFilter-Lite [5] demonstrated that quantised, streaming TSE is practical on phones.

Table 2.1: Comparison of representative separation and extraction systems

System	Task	Speaker cue	Speakers evaluated	Key property
Conv-TasNet [7]	BSS	-	2-3 (separate models)	Learned encoder, TCN masks
SuDoRM-RF [8]	BSS	-	2 / universal	Efficient U-shaped conv blocks
SpeakerBeam [2]	TSE	adaptation layer	2	Speaker-aware network
TD-SpeakerBeam [3]	TSE	multiplicative	2	Time domain, speaker ID loss
VoiceFilter [4]	TSE	d-vector concat	2	Spectrogram masking
VoiceFilter-Lite [5]	TSE	d-vector	2 (+ noise)	Streaming, quantised, on-device ASR
SpEx+ [6]	TSE	multiplicative	2	Shared encoder, multi-scale
This work	TSE	FiLM + gates	**2, 3 and 4 (one model)**	**U-shaped conv, ONNX + INT8, streaming**

2.2 Gaps Identified

- **Fixed two-speaker evaluation.** Most TSE systems are trained and tested only on two-speaker mixtures, although real scenes often contain more talkers. How a TSE model degrades with three or four speakers is rarely reported.

- **Server-side evaluation.** Model size, quantisation effects and streaming latency are seldom measured together with extraction quality.
- **Target confusion.** When the interferer sounds like the target (same gender, similar voice), models extract the wrong speaker; this error is hidden by average scores.
- **Limited metrics.** Many works report only SI-SDR improvement, which does not separate leakage of other speakers from processing artifacts or perceived quality.

This project addresses these gaps by training and evaluating one model for two to four speakers, by measuring the full deployment path from checkpoint to quantised streaming model, by analysing the wrong-speaker errors explicitly, and by reporting eight complementary metrics.

CHAPTER 3
Methodology

3.1 System Architecture

3.1.1 Architecture Overview

The system has two inputs, a mixture waveform y of T samples and an enrollment waveform of the target speaker, and one output, the estimate $\hat{s}$ of the target speech. Figure 3.1 shows the complete network, called U-eConv-TSE. A learned convolutional encoder maps the mixture to a non-negative feature sequence. A speaker encoder turns the enrollment into a 256-dimensional embedding e. A stack of R separator stages, each a U-shaped convolutional block conditioned on e, refines the features; a gated embedding split after every stage produces target-specific features, which a gated aggregation combines across stages. A mask head and a transposed-convolution decoder finally produce the target waveform. All operations are in the time domain; there is no Fourier transform and no phase estimation.

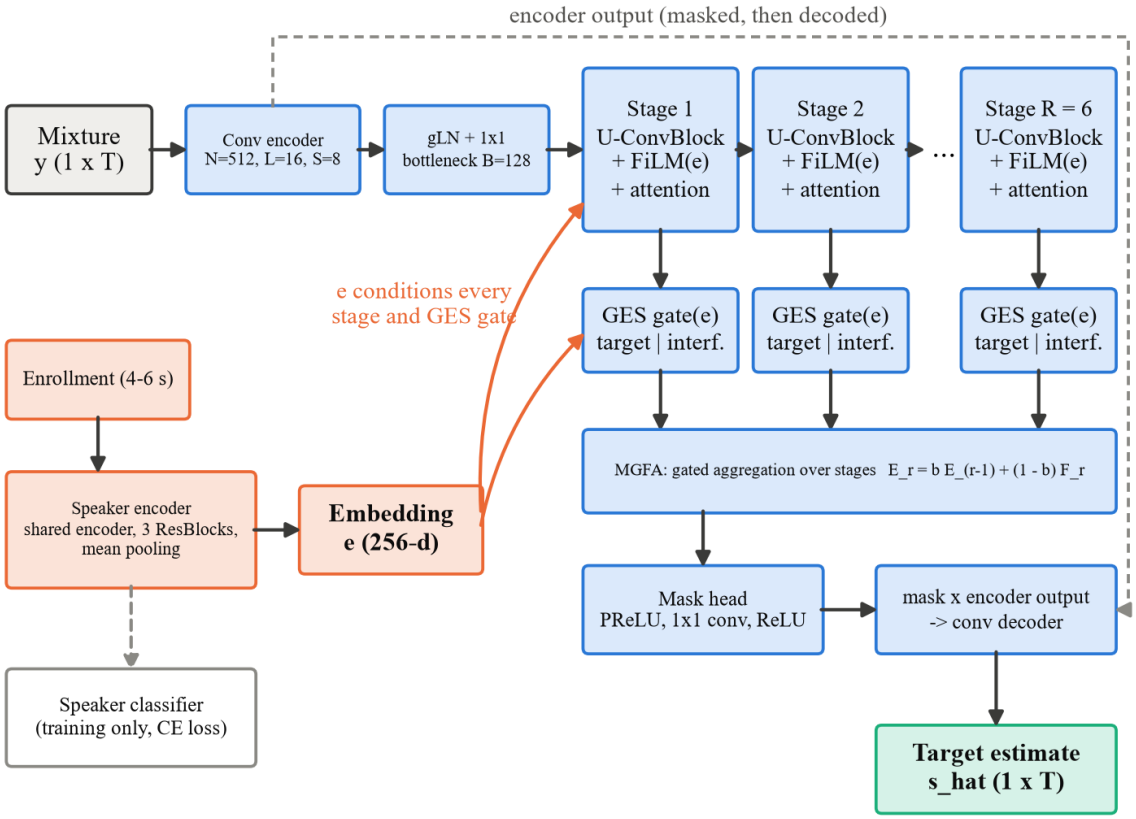


Figure 3.1: Overall architecture of the proposed U-eConv-TSE network (R = 6 stages)

3.1.2 Encoder and Decoder

The encoder is a 1-D convolution with $N = 512$ filters of length $L = 16$ samples (2 ms at 8 kHz) and stride $S = 8$, followed by a ReLU, giving one 512-dimensional frame per millisecond. Global layer normalisation (gLN) and a 1×1 convolution reduce the frames to $B = 128$ channels for the separator. The decoder is the matching transposed convolution; it is applied to the encoder output multiplied by the estimated mask, so the network only has to decide *how much* of each encoder channel belongs to the target at each instant.

3.1.3 Speaker Encoder

The enrollment is passed through the *same* encoder as the mixture, so that the speaker cue and the mixture features live in the same learned representation. A gLN, a 1×1 convolution and three residual blocks (convolution, batch normalisation, PReLU, max-pooling by 3) follow, then mean pooling over time and a linear layer give the 256-dimensional embedding e. During training a linear classifier predicts the speaker identity from e (251 training speakers) and adds a cross-entropy loss, which forces e to be discriminative. The classifier is not used at inference time.

3.1.4 U-ConvBlock Separator with FiLM Conditioning

Each separator stage is a U-ConvBlock (Figure 3.2), adapted from the efficient SuDoRM-RF design [8]. A 1×1 convolution expands the 128 input channels to $H = 512$. Four depthwise-separable convolutions with kernel 5 and stride 2 then halve the time resolution four times (1 ms to 16 ms frames). At the coarsest scale the speaker embedding is injected by FiLM [13]:

$$FiLM(h, e) = \gamma(e) \odot h + \beta(e) \tag{3.1}$$

where γ and β are linear functions of e, initialised so that the layer starts as the identity. An attention module (Section 3.1.6) follows. The block then up-samples back to the input resolution, adding the skip connection of each scale, and a 1×1 convolution and a residual connection return 128 channels. The multi-resolution path lets every stage combine fine detail (pitch harmonics) with long context (syllables and words) at low cost, because most computation happens at reduced time resolution.

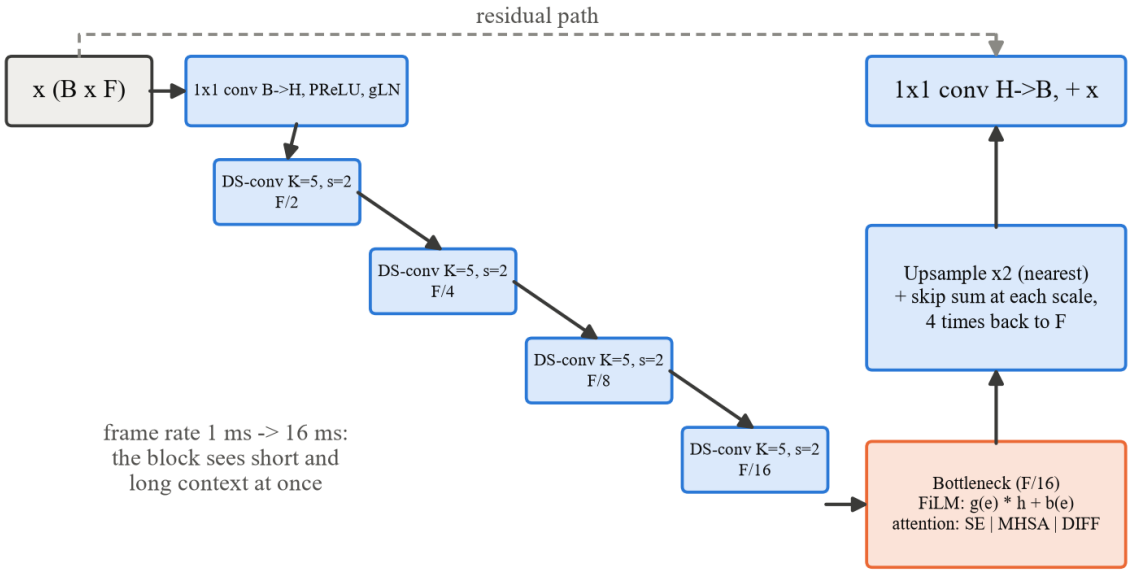


Figure 3.2: One U-ConvBlock separator stage with FiLM conditioning and bottleneck attention

3.1.5 Gated Embedding Split and Multi-stage Gated Feature Aggregation

After every stage a *Gated Embedding Split* (GES) separates the features into a target stream and an interference stream. A 1×1 convolution produces two halves a and g; the gate is conditioned on the speaker, $\mathrm{glu} = a \odot \sigma(g + W e)$, and a second 1×1 convolution with group normalisation splits the result into target features F_t and interference features F_i. The interference stream is decoded only during training, where an auxiliary loss asks it to reconstruct the interfering speech; it is removed for inference. *Multi-stage Gated Feature Aggregation* (MGFA) combines the target features of all stages with a learned per-channel gate:

$$E_r = b \odot E_(r\text{-}1) + (1 - b) \odot F_t,r\,, \quad b = \sigma(w), \quad E_1 = F_t,1 \tag{3.2}$$

so that the mask head sees both early, fine-grained and late, refined features.

3.1.6 Bottleneck Attention: Squeeze-and-Excitation, Multi-Head and Differential

The attention module at the U-block bottleneck is a configurable component (`model.attention`). The baseline is **squeeze-and-excitation (SE)** [14], a channel attention: the features are averaged over time, passed through a two-layer bottleneck MLP and a sigmoid, and the result re-weights the 512 channels. SE gives one weight per channel for the whole segment and cannot relate different time positions.

Two time-attention replacements were implemented. **Multi-head self-attention (MHSA)** [15] treats each 16 ms frame as a token; with layer normalisation (LN), query, key and value projections of width 256 and four heads of width 64,

$$MHSA(h) = h + W_o \cdot concat_k\ softmax(\ Q_k\ K_k^T / \sqrt{d}\ )\ V_k \quad (3.3)$$

Differential attention (DIFF) [16] splits every query and key into two halves and subtracts two attention maps, cancelling attention that both maps assign in common ("attention noise"):

$$DiffAttn(h) = (\ softmax(Q_1K_1^T/\sqrt{d}) - \lambda\ softmax(Q_2K_2^T/\sqrt{d})\ )\ V \quad (3.4)$$

with a learned scalar $\lambda = \exp(\lambda_q1 \cdot \lambda_k1) - \exp(\lambda_q2 \cdot \lambda_k2) + \lambda_init$, where $\lambda_init = 0.8 - 0.6 \cdot \exp(-0.3 \cdot r)$ depends on the stage index r, and a per-head RMS normalisation scaled by $(1 - \lambda_init)$. Two heads of double width are used, so both variants have exactly the same number of parameters. In both, the output projection is initialised to zero, so the block starts as an identity mapping and training is stable. For extraction, the differential form is attractive because the mixture contains speech that resembles the target: attention that responds to all speech alike is exactly the common-mode part that the subtraction removes.

3.2 Dataset and Data Pipeline

3.2.1 Datasets

All audio is 8 kHz, clean (no noise), in the LibriMix *min* mode (mixtures cut to the shortest source). Table 3.1 lists the data. Libri2Mix train-100 provides the 251 training speakers; development and test speakers never appear in training. Libri3Mix test was regenerated exactly from the official LibriMix metadata. Because no public four-speaker set exists, a **Libri4Mix** test set was built by adding a fourth LibriSpeech test-clean speaker (not already present) to every Libri3Mix mixture, at a random level between the quietest and the loudest of the other three, with a fixed random seed.

Table 3.1: Datasets used in this project

Set	Split	Mixtures	TSE samples	Speakers	Use
LibriSpeech train-clean-100	train	dynamic	27,800 per epoch	251	training (dynamic 2-4 spk)
Libri2Mix	dev	3,000	6,000	40	validation (subset 400)
Libri3Mix	dev	3,000	9,000	40	validation (subset 400)
Libri4Mix (ours)	dev	3,000	12,000	40	validation (subset 400)
Libri2Mix	test	3,000	6,000	40	final evaluation
Libri3Mix	test	3,000	9,000	40	final evaluation
Libri4Mix (ours)	test	3,000	12,000	40	final evaluation

Each mixture with n speakers gives n TSE samples, one per possible target speaker.

3.2.2 Enrollment Protocol

LibriMix contains no enrollment recordings, so they are taken from the same speaker's other utterances. For a sample with target speaker S and target utterance U, the enrollment is a clean utterance of S with a different utterance ID, from the same data split. During training a new enrollment is drawn at random every time (4 s crops); for validation and test one enrollment per sample is fixed with a seeded random generator and saved to a table (6 s, centre crop), so every model is evaluated on identical mixture-enrollment pairs.

3.2.3 Dynamic Multi-Speaker Mixing

Instead of the fixed Libri2Mix training mixtures, every training example is mixed on the fly (Figure 3.3). A random 3 s crop of the target utterance is combined with one, two or three interferers drawn from other training speakers, with probabilities 0.40, 0.35 and 0.25 for two, three and four speakers in total. Each source receives an independent level in [−5, +5] dB, and with probability 0.25 an interferer is active for only part of the segment (at least half), which imitates people starting and stopping. The model therefore never sees the same mixture twice and learns to handle a variable number of talkers.

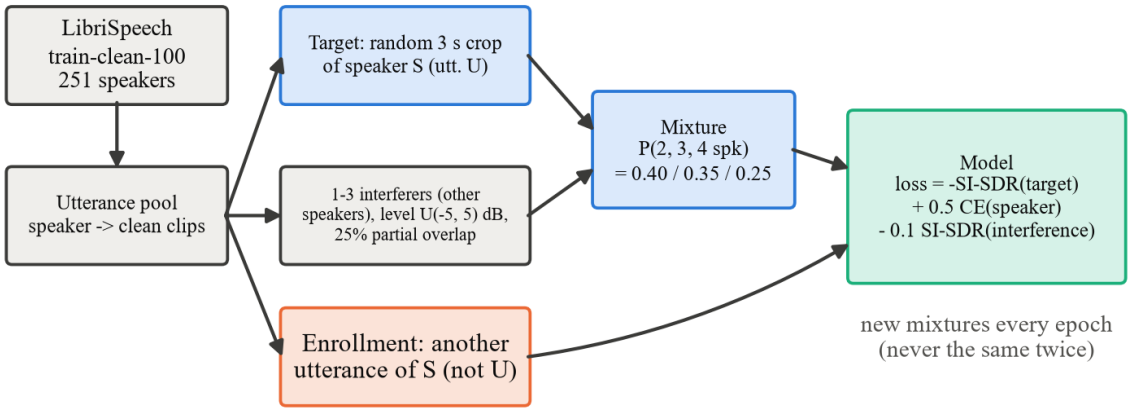


Figure 3.3: Training data pipeline with dynamic 2-4 speaker mixing

3.3 Training Objective and Procedure

The scale-invariant signal-to-distortion ratio (SI-SDR) [21] between the estimate $\hat{s}$ and the reference s is

$$\text{SI-SDR}(\hat{s}, s) = 10 \log_{10}(\|\alpha\, s\|^2 / \|\alpha\, s - \hat{s}\|^2), \quad \alpha = \langle \hat{s}, s \rangle / \|s\|^2 \tag{3.5}$$

after removing the mean of both signals. The training loss combines the negative SI-SDR of the target, the speaker-classification cross-entropy (CE) and the negative SI-SDR of the auxiliary interference estimate $\hat{s}_i$ against the true interference $y - s$:

$$\mathcal{L} = -SI\text{-}SDR(\hat{s}, s) + 0.5 \cdot CE(e) - 0.1 \cdot SI\text{-}SDR(\hat{s}_i, y - s) \quad (3.6)$$

Table 3.2: Model and training hyper-parameters

Parameter	Value
Encoder N / L / S	512 / 16 / 8
Bottleneck B, U-block width H	128, 512
U-block depth Q, kernel K	4, 5
Stages R	6 (proposed), 3 (compact)
Speaker embedding	256-d, ResNet encoder [256, 512, 512]
Segment / enrollment length	3 s / 4 s (train), full / 6 s (test)
Batch size, optimiser	8, Adam [35]
Learning rate	1e-3, halved after 3 epochs without improvement
Epochs	80 (from scratch)
Gradient clipping	5.0
Loss weights	λ_CE = 0.5, λ_int = 0.1
Model selection	best mean validation SI-SDRi over 2/3/4-speaker dev subsets
Training hardware / time	NVIDIA RTX 5090, about 4 min per epoch (about 5.5 h)

Three models are compared in Chapter 4. The **2-speaker model** (R = 3) is trained on the fixed Libri2Mix training mixtures. The **fine-tuned model** continues training the 2-speaker model for 25 epochs with dynamic 2-4 speaker mixing at a learning rate of 2.5e-4. The **proposed model** (R = 6) is trained from scratch for 80 epochs with dynamic 2-4 speaker mixing.

3.4 Evaluation Metrics

Every test sample is scored with the following metrics; the *improvement* versions (suffix i) subtract the value obtained by the unprocessed mixture.

- **SI-SDR / SI-SDRi** [21]: overall extraction quality, insensitive to output level. The main metric.
- **SNR / SNRi**: ordinary signal-to-noise ratio after matching the estimate's energy to the reference. Because the model is trained with a scale-invariant loss its output level is arbitrary, so plain SNR would mostly measure loudness; energy matching keeps leakage and distortion in the error.
- **SDR, SIR, SAR** [22]: BSS-eval decomposition. The estimate is projected (512-tap filters) onto the target alone and onto all sources (each interfering speaker separately). SIR measures leakage of the other speakers, SAR the artifacts introduced by processing, SDR both.

- **PESQ** [23]: perceived speech quality, ITU-T P.862 narrow-band, on a MOS-like scale (about 1 to 4.5).
- **STOI** [24]: predicted intelligibility, between 0 and 1.
- **NSR** (negative SI-SDRi rate): the share of samples where the output is *worse* than the mixture, which in practice means the wrong speaker was extracted. Also reported: the share below 5 dB.

The BSS-eval metrics were implemented directly with NumPy and SciPy (Cholesky solution of the block-Toeplitz projection), because the available library failed with NumPy 2. The implementation was verified against the reference *mir_eval* implementation on real three-speaker mixtures (agreement within 0.01 dB) and against the SDR values reported earlier in the project (mean difference below 0.001 dB), and is about 20 times faster.

3.5 Edge Deployment Pipeline

3.5.1 Graph Split and ONNX Export

At inference time the enrollment is processed only once per target speaker, while the mixture is processed continuously. The network is therefore split into two graphs (Figure 3.4): an **embedder** (enrollment → 256-d embedding) and an **extractor** (mixture window + embedding → target window). The training-only parts (speaker classifier, interference decoder) are removed. Both graphs are exported to ONNX (operator set 17) with a dynamic time axis, so any window length can be processed. The exported graphs were checked against PyTorch at lengths different from the export length.

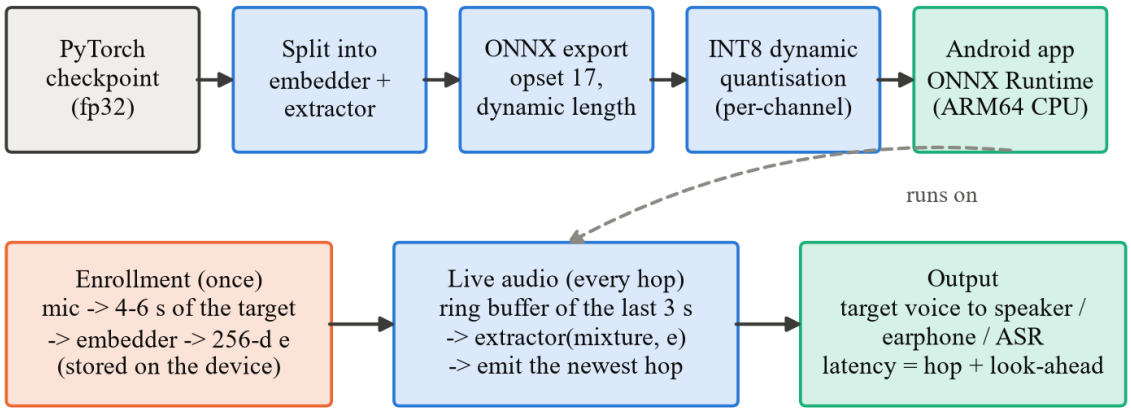


Figure 3.4: Edge deployment pipeline: from the trained checkpoint to streaming inference on the device

3.5.2 Quantisation

Two reduced-precision versions are produced: **FP16**, where weights and activations use 16-bit floats, and **INT8**, using ONNX Runtime dynamic quantisation with per-channel 8-bit weights; activations are quantised on the fly. INT8 reduces the stored model to about a quarter of its FP32 size and allows integer arithmetic on edge CPUs. No retraining or calibration data are needed.

3.5.3 Streaming Inference

The separator is non-causal: global normalisation and two-sided convolutions look at the whole input window. Real-time operation therefore uses a sliding window (Figure 3.5). The device keeps the last $W = 3$ s of audio in a ring buffer; every *hop* it runs the extractor on the buffer and outputs the hop that ends *look-ahead* seconds before the newest sample. The algorithmic latency is hop + look-ahead, and the extractor must process one window in less than one hop of wall-clock time.

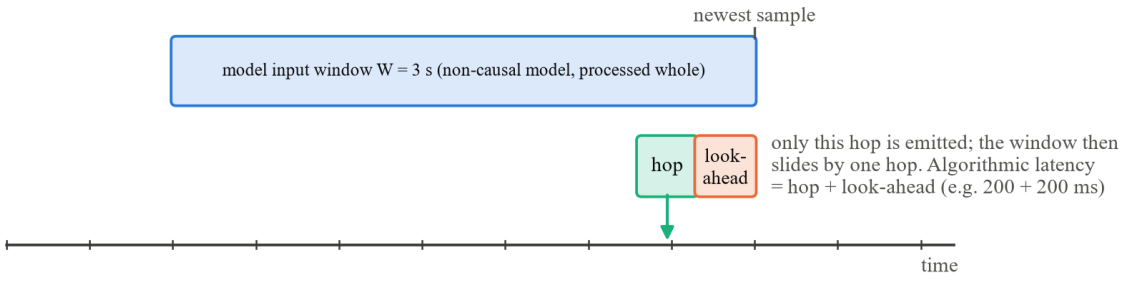


Figure 3.5: Sliding-window streaming: only the newest hop of each 3 s window is output

3.5.4 Target Edge Platform: Android Smartphone Application

The edge platform is an Android smartphone, the most widely available edge device with a microphone, an ARM CPU and a battery. A native Android application, *TSE Edge*, was developed for it (Figure 3.6). The application bundles the INT8 embedder and extractor of both models (R = 3: 8.6 MB, R = 6: 13.6 MB) and the ONNX Runtime Android library (version 1.20, CPU execution provider, ARM64), and it runs the complete extraction on the phone. It requests only the microphone permission and has **no internet permission**, so no audio or result can leave the device; it works in airplane mode.

- **Inference engine** (`TseEngine`, Java): loads the two ONNX graphs from the application package, computes the speaker embedding once per enrollment, runs the extractor on whole recordings or on 3 s streaming windows, and implements the

same sliding-window streaming (least-squares level matching, 20 ms crossfade) as the PC implementation.

- **Audio**: recording with the Android `AudioRecord` API at 8 kHz mono (16 kHz with a 31-tap low-pass decimation filter where 8 kHz is unsupported), playback with `AudioTrack`.
- **User flow**: (1) record the user's voice alone for 6 s (enrollment); (2) record or receive a multi-talker mixture; (3) the phone extracts and plays back only the enrolled voice.
- **Built-in benchmark**: measures model load time, enrollment time, time per 3 s window and real-time factor for 1 and 4 CPU threads, the SI-SDRi on bundled test mixtures (compared with the PC result for the same files), and two streaming latency settings; the results can be shared from the phone.
- **Verification**: the engine was first run on a desktop JVM with the same ONNX Runtime version and produced exactly the same SI-SDRi as the Python implementation on all bundled test mixtures (difference 0.000 dB).

The measurements in Section 4.3 were made on a Samsung SM-S731B smartphone (Exynos 2400 SoC, 10 cores CPU, 7.0 GB RAM, Android 17). The application targets Android 7.0 (API 24) and newer on 64-bit and 32-bit ARM phones.

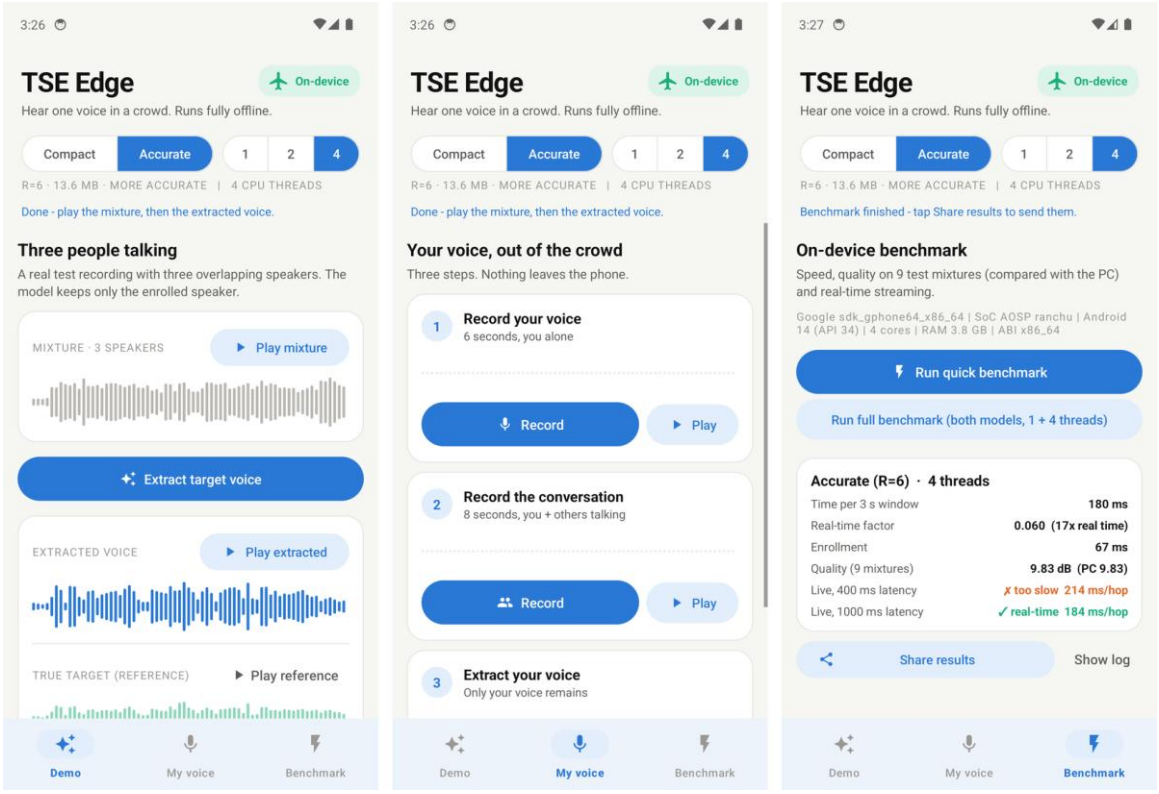


Figure 3.6: The TSE Edge Android application: demo with a 3-speaker test mixture (left), extraction of the user's own voice (centre) and on-device benchmark (right)

3.6 Software and Hardware Setup

Table 3.3: Development environment

Item	Version / specification
Training and evaluation host	Intel Core Ultra 9 285K (24 cores), NVIDIA GeForce RTX 5090 (32 GB)
Operating system	Windows 11
Language / framework	Python 3.13, PyTorch 2.11 (CUDA 12.8)
Export and runtime	ONNX 1.23, ONNX Runtime 1.31 (CPU execution provider)
Metrics	pesq (ITU-T P.862), pystoi, own NumPy/SciPy BSS-eval
Testing	pytest: 81 automated tests (models, data, metrics, training, streaming)
Edge device	Samsung SM-S731B smartphone (Exynos 2400 SoC, 10 cores CPU, 7.0 GB RAM, Android 17); ONNX Runtime Android 1.20 (CPU, ARM64)
Android application	Java, Android API 24-34, built with the Android SDK 34 command-line tools (aapt2, d8, apksigner); APK 26.8 MB including models

CHAPTER 4
Results and Discussion

All numbers in this chapter come from complete runs of the project's evaluation scripts on the full test sets (6,000, 9,000 and 12,000 samples for two, three and four speakers), with the fixed enrollment tables. The edge measurements in Section 4.3 were made on a smartphone by the TSE Edge application itself.

4.1 Results and Comparison

4.1.1 Extraction Quality versus Number of Speakers

Table 4.1: Extraction quality on the 2-, 3- and 4-speaker test sets (higher SI-SDRi / SDRi is better; lower NSR is better)

Test set	Model	SI-SDRi (dB)	median	SDRi (dB)	NSR	< 5 dB
2 speakers	R=3, 2-spk training	9.61	10.98	10.12	5.6%	12.0%
2 speakers	R=3, fine-tuned 2-4 spk	8.73	10.48	9.41	8.8%	17.0%
2 speakers	**R=6, 2-4 spk (proposed)**	9.49	11.11	10.15	7.5%	13.0%
3 speakers	R=3, 2-spk training	4.84	5.36	4.97	7.1%	46.1%
3 speakers	R=3, fine-tuned 2-4 spk	7.19	8.49	7.78	8.3%	22.4%
3 speakers	**R=6, 2-4 spk (proposed)**	8.13	9.60	8.76	7.8%	17.6%
4 speakers *	R=3, 2-spk training	3.13	3.40	3.20	8.9%	76.9%
4 speakers *	R=3, fine-tuned 2-4 spk	6.06	7.05	6.63	8.5%	30.3%
4 speakers *	**R=6, 2-4 spk (proposed)**	7.00	8.33	7.61	8.5%	23.9%

** Libri4Mix is this project's four-speaker extension of Libri3Mix, not an official benchmark.*

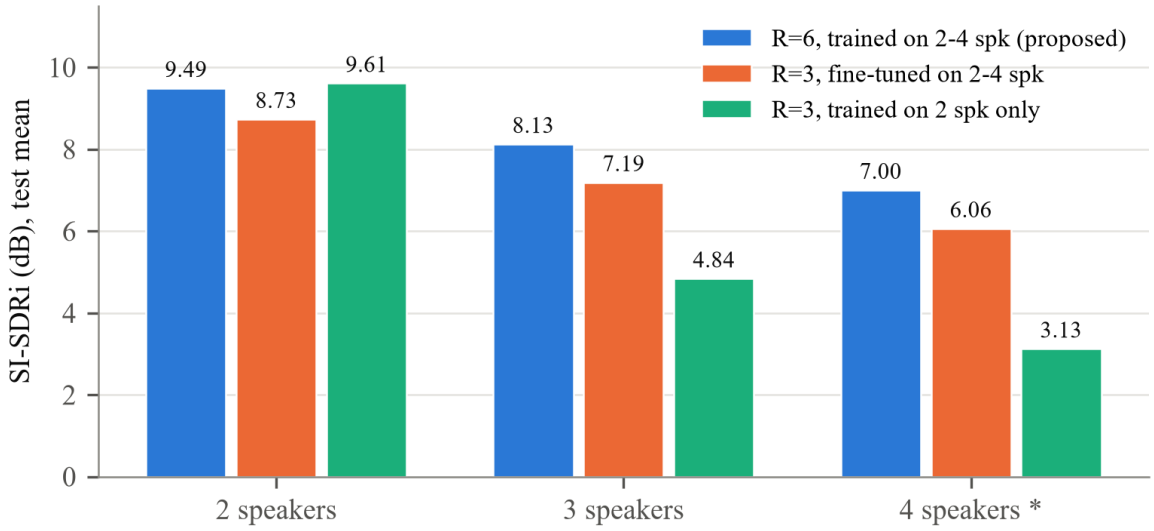


Figure 4.1: Mean SI-SDRi on the test sets as the number of speakers grows

Table 4.1 and Figure 4.1 show the central result. The model trained only on two-speaker mixtures works well on two speakers (9.61 dB) but loses about half of its improvement

when a third speaker is added (4.84 dB) and more with a fourth (3.13 dB); 76.9% of its four-speaker outputs improve by less than 5 dB. Training with dynamic 2-4 speaker mixing removes most of this loss. The proposed R = 6 model reaches 8.13 dB on three speakers (+3.29 dB) and 7.00 dB on four (+3.88 dB), while keeping two-speaker quality within 0.12 dB of the specialised model. A single model can therefore serve scenes with an unknown number of talkers.

Fine-tuning the existing two-speaker model is a cheaper route: it recovers most of the multi-speaker quality in 25 epochs, but it costs 0.9 dB on two speakers and stays about 1 dB behind the proposed model on three and four speakers. Two things differ between the fine-tuned and the proposed model (depth 3 → 6 and 25 fine-tuning epochs → 80 epochs from scratch), so the gain cannot be attributed to depth alone.

4.1.2 Perceptual Quality, Interference and Artifacts

Table 4.2: Signal and perceptual metrics on the test sets (mixture → output for PESQ and STOI)

Test set	Model	SNRi (dB)	SIRi (dB)	SAR (dB)	PESQ	STOI
2 speakers	R=3, 2-spk training	-8.46	17.26	12.32	1.60 → 2.25	0.713 → 0.858
2 speakers	R=3, fine-tuned 2-4 spk	-8.42	16.74	11.44	1.60 → 2.19	0.713 → 0.841
2 speakers	**R=6, 2-4 spk (proposed)**	7.80	17.61	12.22	1.60 → 2.26	0.713 → 0.854
3 speakers	R=3, 2-spk training	-6.19	7.33	8.77	1.37 → 1.58	0.592 → 0.703
3 speakers	R=3, fine-tuned 2-4 spk	-6.36	12.85	8.02	1.37 → 1.67	0.592 → 0.734
3 speakers	**R=6, 2-4 spk (proposed)**	5.58	14.39	8.77	1.37 → 1.75	0.592 → 0.756
4 speakers	R=3, 2-spk training	-5.07	4.69	7.89	1.30 → 1.40	0.528 → 0.602
4 speakers	R=3, fine-tuned 2-4 spk	-5.36	10.49	6.53	1.30 → 1.48	0.528 → 0.653
4 speakers	**R=6, 2-4 spk (proposed)**	4.16	11.93	7.19	1.30 → 1.54	0.528 → 0.679

The BSS-eval decomposition explains *where* the improvement comes from. For the proposed model on three speakers the interference is suppressed by 14.39 dB (SIRi), while the artifacts stay at an SAR of 8.77 dB; the output is therefore limited more by processing artifacts than by leftover speakers. Perceived quality rises from a PESQ of 1.37 for the mixture to 1.75, and predicted intelligibility from 0.592 to 0.756. The two-speaker-only model suppresses the interferers much less on three and four speakers (SIRi 7.33 and 4.69 dB, against 14.39 and 11.93 dB for the proposed model), which confirms that its failure mode is leakage of the extra talkers.

4.1.3 Training Behaviour

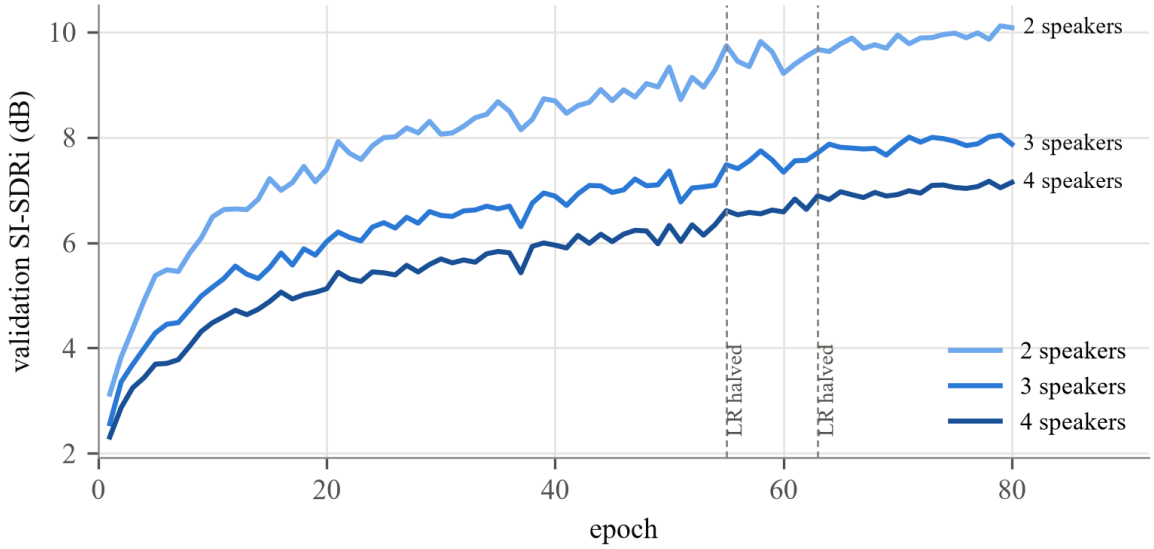


Figure 4.2: Validation SI-SDRi of the proposed model during training, for each number of speakers

Figure 4.2 shows that the three validation curves rise together and stay ordered by difficulty throughout training. The learning rate was halved twice, after which the curves still rise slowly; the best checkpoint was reached at epoch 79 of 80, so the model had not fully converged and a longer schedule would likely improve it further.

4.1.4 Error Analysis

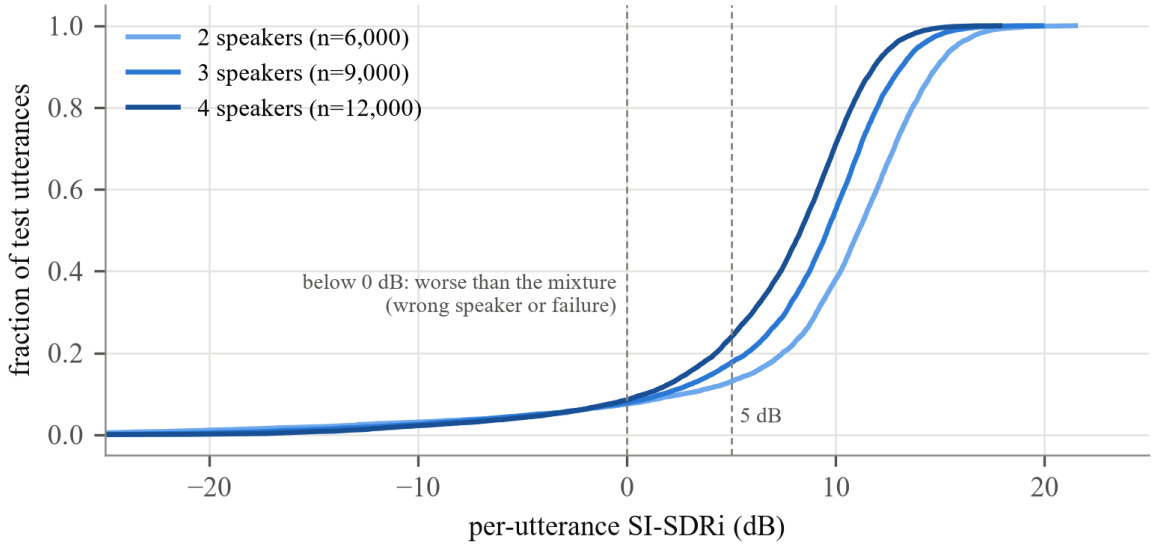


Figure 4.3: Cumulative distribution of per-utterance SI-SDRi of the proposed model

Average scores hide two different kinds of error (Figure 4.3). Most outputs are good: the median SI-SDRi is 11.11, 9.60 and 8.33 dB for two, three and four speakers. A small group of outputs, however, lies below 0 dB (NSR 7.5%, 7.8% and 8.5%): in these cases the model

followed the wrong speaker. Unlike leakage, this error did not decrease with more training data or more stages.

Table 4.3: SI-SDRi (dB) of the proposed model by speaker gender (same gender: at least one interferer has the target's gender)

Test set	same gender	n	different gender	n
2 speakers	8.55	2,972	10.42	3,028
3 speakers	7.65	6,714	9.52	2,286
4 speakers	6.75	10,468	8.73	1,532

Table 4.3 shows that wrong-speaker errors concentrate where voices are similar. When an interferer has the same gender as the target, SI-SDRi is 1.9-2.0 dB lower than when all interferers have the other gender. A finer analysis with a pretrained speaker-verification model (ECAPA-TDNN [18]) measured the similarity of every target-interferer pair: on two speakers the wrong-speaker rate rises from 7.0% for the least similar fifth of pairs to 10.2% for the most similar fifth. The weak point of the system is therefore the speaker cue, not the separation capacity.

4.1.5 Qualitative Example

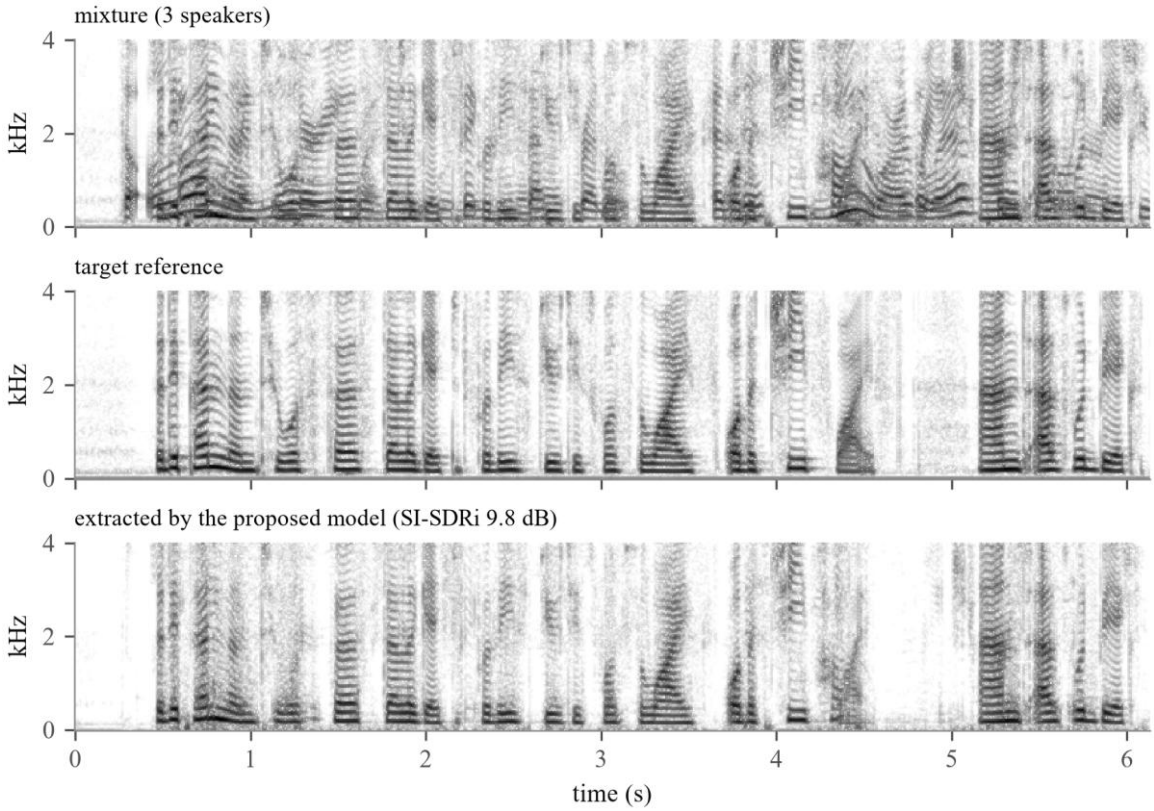


Figure 4.4: Spectrograms of a three-speaker test mixture, the target reference and the extracted speech (an utterance at the median SI-SDRi of the test set)

Figure 4.4 shows a three-speaker test mixture chosen automatically as the utterance whose SI-SDRi is closest to the test median (9.76 dB), so it is representative rather than selected. The harmonics of the two interfering speakers that fill the gaps of the target are removed, while the target's harmonic structure and its silences (for example around 4.6-5.1 s) are preserved.

4.2 Attention Study: Multi-Head and Differential Attention

To test whether attention over time helps the separator, the SE channel attention was replaced by multi-head self-attention (MHSA) or differential attention (DIFF), with everything else identical: same data, recipe, random seed and number of epochs as the proposed model. Only the attention module changes. Both attention variants add 2.76 M parameters (16.36 M against 13.60 M in total) and about 4-7% training time per step.

The two attention runs were stopped after **17 (MHSA) and 15 (DIFF) of the planned 80 epochs** (see below). The comparison is therefore made at matched epochs: all three models are compared at epoch 15, using the per-epoch validation log of the proposed model.

Table 4.4: Validation results at epoch 15 with identical training (SI-SDRi in dB; mean = average over 2/3/4 speakers)

Bottleneck attention	2 spk	3 spk	4 spk	mean	best ≤ ep. 15	NSR
SE (proposed model, channel attention)	7.22	5.53	4.88	5.88	5.88	9.6%
MHSA (multi-head self-attention)	5.07	4.28	3.61	4.32	4.32	8.6%
DIFF (differential attention)	5.47	4.37	3.82	4.55	4.55	10.6%

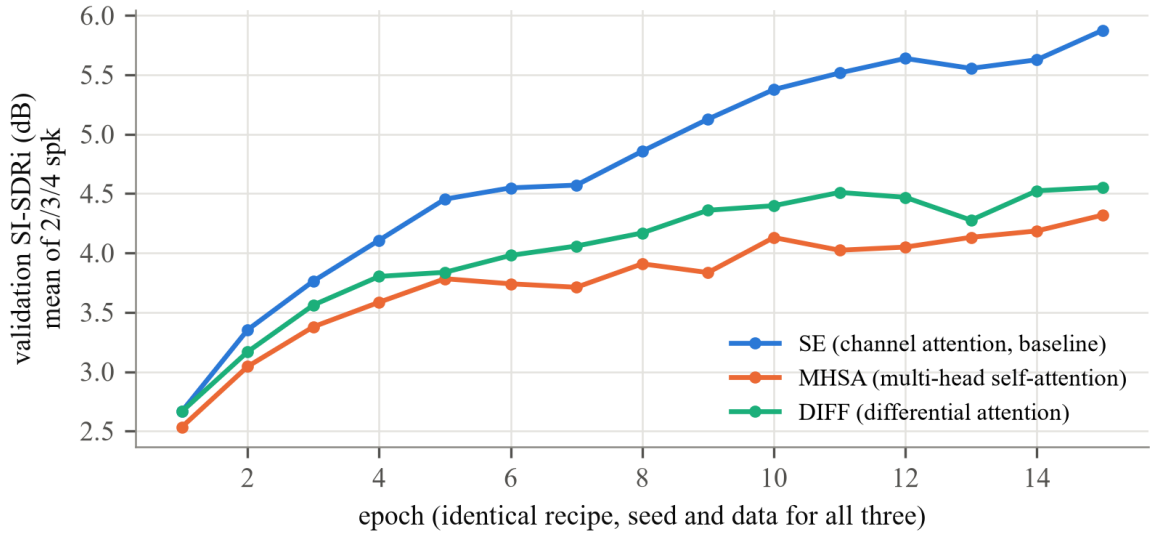


Figure 4.5: Validation SI-SDRi per epoch for SE, MHSA and DIFF bottleneck attention (identical recipe; first 15 epochs)

At epoch 15 the mean validation SI-SDRi is: SE 5.88 dB, DIFF 4.55 dB, MHSA 4.32 dB. SE is ahead by 1.32 dB, far more than the typical epoch-to-epoch change of the validation score (0.21 dB), so the difference is real. Two observations are already useful for the edge goal. First, the attention variants add 20% parameters but little compute, because attention runs at the 16 ms bottleneck resolution where sequences are short. Second, differential attention trains as stably as standard attention, and lags the baseline less than multi-head attention. Because the gap to the SE baseline widened steadily instead of closing (MHSA from -0.13 dB at epoch 1 to -1.59 dB at epoch 17; DIFF from 0.00 dB at epoch 1 to -1.32 dB at epoch 15), and finishing the 80 epochs would have taken about nine more hours, both runs were stopped. With the same recipe and learning rate as the convolutional baseline, replacing SE by time attention therefore did not help. A likely reason is that attention layers usually need a learning-rate warm-up and a lower learning rate than this recipe (1e-3 from the first step) provides; this is left for future work. The checkpoints are kept, so the runs can be resumed.

4.3 Edge Deployment Results

4.3.1 Model Size and Complexity

Table 4.5: Size and compute of the deployed graphs

Model	Embedder params (M)	Extractor params (M)	Extractor GFLOPs per s of audio	Embedder GFLOPs (once)	FP32 size (MB)	INT8 size (MB)
R=3 (compact)	3.55	4.98	2.88	9.15	32.7	8.6
R=6 (proposed)	3.55	9.82	5.46	9.15	51.2	13.6

FLOPs count a multiply-add as two operations. Sizes are the ONNX files of both graphs together.

Splitting the network pays off: the embedder costs 9.15 GFLOPs but runs only once per enrolled speaker, while the extractor costs 5.46 GFLOPs per second of audio for R = 6 and about half that for R = 3. INT8 quantisation cuts the stored model to about a quarter of its size, which fits comfortably in the memory of embedded processors. The exported ONNX graphs reproduce the PyTorch outputs: the SI-SDR between the ONNX and the PyTorch output was 53.4 dB for R = 6 and 58.6 dB for R = 3, i.e. numerically identical for audio purposes.

4.3.2 Effect of Quantisation on Quality

Table 4.6: SI-SDRi (dB) of the reduced-precision models (change against FP32 in brackets)

Model	Variant	2 spk	3 spk	4 spk	agreement with FP32 (dB)
R=6	PyTorch FP32 (reference)	11.08 (+0.00)	8.03 (+0.00)	7.23 (+0.00)	-
R=6	PyTorch FP16	fails (overflow)	fails (overflow)	fails (overflow)	-
R=6	ONNX Runtime FP32	11.08 (+0.00)	8.03 (-0.00)	7.23 (-0.00)	58.7
R=6	ONNX Runtime INT8	10.94 (-0.13)	8.05 (+0.02)	7.16 (-0.07)	25.7
R=3	PyTorch FP32 (reference)	9.75 (+0.00)	6.84 (+0.00)	6.49 (+0.00)	-
R=3	PyTorch FP16	fails (overflow)	fails (overflow)	fails (overflow)	-
R=3	ONNX Runtime FP32	9.75 (-0.00)	6.84 (-0.00)	6.49 (+0.00)	60.5
R=3	ONNX Runtime INT8	9.69 (-0.06)	6.75 (-0.09)	6.45 (-0.04)	24.4

Fixed, seeded subsets of 150 test samples per set (same mixtures and enrollments for all variants). Agreement = median SI-SDR between the variant's output and the FP32 output, 3 speakers.

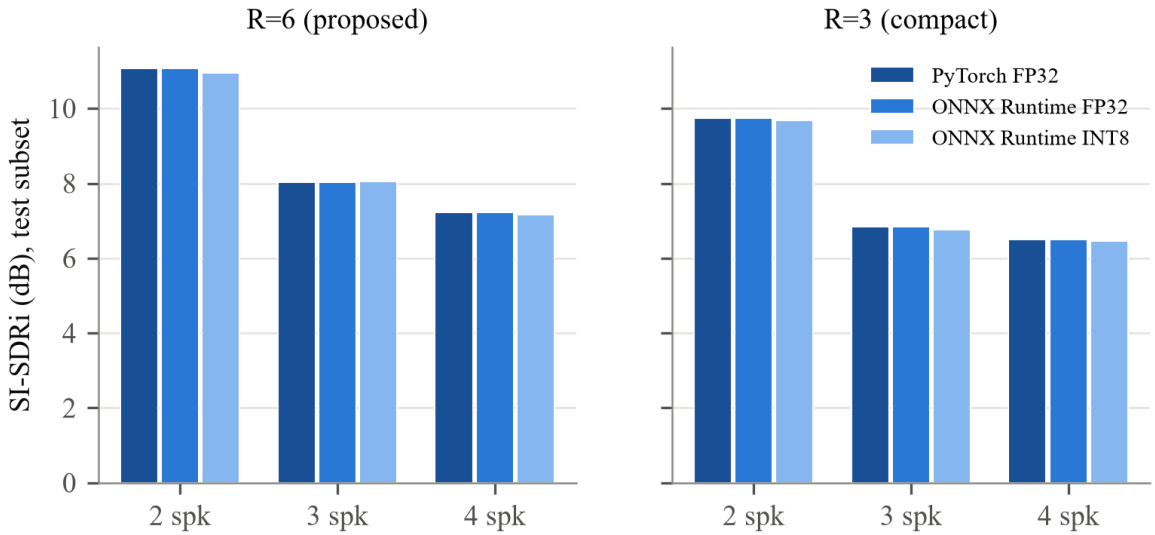


Figure 4.6: SI-SDRi of the FP32 and INT8 versions on 2/3/4-speaker test subsets

The exported ONNX FP32 graph reproduces the PyTorch reference exactly. INT8 quantisation changes SI-SDRi on two, three and four speakers by -0.13, +0.02, -0.07 dB for the proposed R = 6 model; and by -0.06, -0.09, -0.04 dB for the compact R = 3 model, a negligible loss for a four-fold smaller model, which makes INT8 the deployment format. Pure FP16 inference, in contrast, **fails** and produces non-finite outputs. Tracing the FP16 forward pass located the first non-finite values in the gated embedding split (GES): its group normalisation computes statistics from activations that reach about 2×10^4 in the FP32 model, and their squares (about 4×10^8) far exceed the largest FP16 value (65,504). The network was trained in FP32 with a scale-invariant loss, which leaves internal levels unconstrained. FP16 deployment would need the normalisation layers kept in FP32 or precision-aware fine-tuning. This is an instructive edge-deployment result: 8-bit integer

weights with FP32 activations are safer for this model than halving the floating-point precision.

4.3.3 Measurements on the Smartphone

The TSE Edge application was installed on a Samsung SM-S731B smartphone (Exynos 2400 SoC, 10 cores CPU, 7.0 GB RAM, Android 17) and run in airplane mode. All numbers below were measured by the application itself on the phone's CPU (ONNX Runtime, INT8, 4 threads) and are reported unchanged.

Table 4.7: On-device measurements on the Samsung SM-S731B (INT8, 4 CPU threads)

Quantity	Accurate model (R=6)	Compact model (R=3)
Model load (both graphs)	292 ms	117 ms
Enrollment, 6 s of speech (once)	80 ms	79 ms
Time per 3 s window (median / p90)	228 / 231 ms	116 / 119 ms
Real-time factor (3 s window)	0.076 (13x faster than real time)	0.039 (26x faster than real time)
Mean SI-SDRi, 9 bundled test mixtures	9.84 dB (PC 9.83 dB)	9.08 dB (PC 9.08 dB)
Largest per-mixture difference to PC	0.17 dB	0.07 dB
Java heap of the application	30 MB	26 MB
Model files in the application	13.6 MB	8.6 MB

Table 4.7 is the main deployment result. The proposed model processes a 3 s window in 228 ms, a real-time factor of 0.076, i.e. about 13 times faster than real time, and the once-per-speaker enrollment takes 80 ms. Extraction quality on the phone (9.84 dB on the nine bundled test mixtures) equals the PC result for the same INT8 model (9.83 dB). The individual mixtures differ by up to 0.17 dB in both directions, most likely because the ARM build of ONNX Runtime uses different integer matrix-multiplication kernels (and activation rounding) than the x86 build; the differences average out.

In interactive use the application extracted the user's voice from 8 s recordings of the phone microphone in 0.61-0.66 s (real-time factor 0.077-0.082), with an enrollment time of 75-140 ms. From the user's point of view the result is available a fraction of a second after the recording ends.

Table 4.8: On-device speed for one and four CPU threads (INT8)

Model	Threads	Load	Enrollment	3 s window	Real-time factor	SI-SDRi (9 mix.)
Compact (R=3)	1	225 ms	104 ms	145 ms	0.048 (21x)	9.08 dB
Compact (R=3)	4	117 ms	79 ms	116 ms	0.039 (26x)	9.08 dB
Accurate (R=6)	1	285 ms	97 ms	275 ms	0.092 (11x)	9.84 dB
Accurate (R=6)	4	292 ms	80 ms	228 ms	0.076 (13x)	9.84 dB

Table 4.8 shows that even a **single CPU core** is enough for both models: one core processes a 3 s window in 145 ms (compact) and 275 ms (accurate). Four threads shorten this by only 20% and 17%, because the model processes one short window at a time and the work per operation is small; on a phone, a single thread is likely the more power-efficient choice. The quality does not depend on the number of threads.

4.3.4 Real-Time Streaming on the Smartphone

Table 4.9: Sliding-window streaming on the phone (3 s window, 3-speaker test mixture)

Model	Thr.	Hop + look-ahead	Latency	Compute / hop	CPU load	Real-time	SI-SDRi
Compact	1	200 + 200 ms	400 ms	142 ms	71%	yes	6.82 dB
Compact	1	500 + 500 ms	1000 ms	142 ms	28%	yes	6.58 dB
Compact	4	200 + 200 ms	400 ms	116 ms	58%	yes	6.82 dB
Compact	4	500 + 500 ms	1000 ms	117 ms	23%	yes	6.58 dB
Accurate	1	200 + 200 ms	400 ms	285 ms	142%	no	7.35 dB
Accurate	1	500 + 500 ms	1000 ms	273 ms	55%	yes	8.07 dB
Accurate	4	200 + 200 ms	400 ms	223 ms	111%	no	7.35 dB
Accurate	4	500 + 500 ms	1000 ms	225 ms	45%	yes	8.07 dB

Real-time: 90% of the hops are computed within the hop duration. CPU load = compute time / hop duration. Whole-utterance SI-SDRi on this mixture: compact 6.76 dB, accurate 8.82 dB.

Table 4.9 shows the streaming behaviour on the phone. Every hop requires one pass over the 3 s window. The **compact model streams in real time with 400 ms of latency even on a single core** (142 ms of computation for every 200 ms of audio, 71% CPU load), with an SI-SDRi of 6.82 dB on this mixture. The accurate model needs about 223 ms per pass, slightly more than a 200 ms hop, so it is run with a 500 ms hop: it then streams in real time with 1 s of latency at 45% CPU load and reaches 8.07 dB (whole-utterance 8.82 dB). The two models therefore offer a clear choice on the device: the compact model for conversation-like latency, the accurate model where one second of delay is acceptable (for example captions, recordings or voice notes). A causal model (Chapter 5) would remove the window overhead and allow lower latency for both.

4.3.5 Streaming Latency versus Quality (Full Test Subsets)

Table 4.10: Streaming quality (SI-SDRi, dB) for different latencies (3 s window)

Model	Latency	hop / look-ahead (ms)	2 spk	3 spk	4 spk	GPU ms per hop
R=6	100 ms	100 / 0	7.76	6.55	5.52	69.8
R=6	300 ms	200 / 100	8.82	7.49	6.32	70.0
R=6	400 ms	200 / 200	9.18	7.90	6.51	71.0

Model	Latency	hop / look-ahead (ms)	2 spk	3 spk	4 spk	GPU ms per hop
R=6	500 ms	250 / 250	9.29	8.05	6.60	69.4
R=6	500 ms	200 / 300	9.25	8.06	6.63	70.1
R=6	1000 ms	500 / 500	9.53	8.29	6.88	69.1
R=6	offline	-	9.83	8.63	7.22	-
R=3	100 ms	100 / 0	7.47	6.29	5.26	9.6
R=3	300 ms	200 / 100	8.53	6.96	5.78	9.6
R=3	400 ms	200 / 200	8.72	7.30	5.92	9.6
R=3	500 ms	250 / 250	8.77	7.32	5.97	9.4
R=3	500 ms	200 / 300	8.83	7.36	6.00	9.4
R=3	1000 ms	500 / 500	9.00	7.44	6.22	9.5
R=3	offline	-	9.21	7.77	6.39	-

Fixed, seeded subsets of 200 test samples per set; the same samples for every setting.

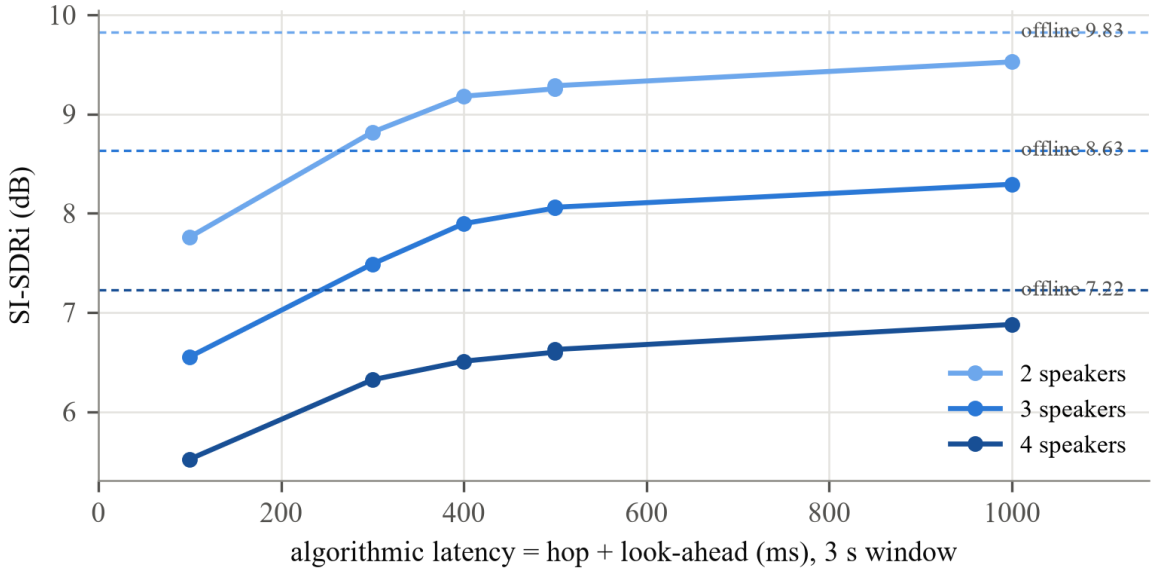


Figure 4.7: Streaming quality of the proposed model against algorithmic latency; dashed lines: whole-utterance (offline) processing

The table extends the single-utterance streaming test of the phone to fixed subsets of the 2-, 3- and 4-speaker test sets (computed on the GPU, so only the quality columns are relevant). The quality loss against whole-utterance processing shrinks as the look-ahead grows, because each emitted hop then has more future context (offline 3-speaker SI-SDRi 8.63 dB).

4.4 Comparison of Existing Systems and the Proposed System

Published TSE systems are trained and evaluated under different conditions (sample rate, data, enrollment selection, number of speakers), so their numbers cannot be compared directly with ours. The following table therefore compares the systems by capability.

Table 4.11: Capabilities of existing systems and the proposed system

Capability	Typical TSE systems [2]-[6]	Proposed system
Number of speakers	trained and tested on 2	one model for 2, 3 and 4
Speaker conditioning	one fusion point	FiLM in every stage + speaker-gated GES
Metrics	mostly SI-SDRi	SI-SDRi, SNRi, SDRi, SIRi, SAR, PESQ, STOI, NSR
Error analysis	average scores	wrong-speaker rate, gender and similarity breakdown
Deployment	server GPU (VoiceFilter-Lite [5]: on-device)	ONNX + INT8; offline Android app, real-time on a smartphone CPU
Model size	typically 5-20 M parameters	13.4 M (R=6), 8.5 M (R=3)

4.5 Negative Results

Two attempts to strengthen the speaker cue did not work, and both are reported because they explain the design choices.

- **Pretrained ECAPA-TDNN as the speaker encoder.** Replacing the jointly trained encoder with a frozen ECAPA-TDNN (speaker-verification equal error rate 2.0% against 6.1% for our encoder) made extraction much worse: after 12 epochs the validation SI-SDRi was 2.27 dB with a 29% wrong-speaker rate, against 5.64 dB and 8% for the jointly trained encoder at the same point, and the train-validation gap was 4.4 dB against 1.9 dB. With only 251 training speakers the separator memorised the very distinct fixed embeddings instead of learning to follow an unseen voice. A better speaker-verification embedding is not automatically a better extraction cue.
- **ECAPA-TDNN as a training judge.** Fine-tuning the proposed model with an extra loss that rewards outputs that sound like the target and unlike each interferer (to an ECAPA judge), together with hard training pairs of similar voices, did not beat a control run with the same hard pairs but no judge loss: the judge run reached a best validation SI-SDRi of 8.47 dB when it was stopped after 21 of its 25 epochs, against 8.54 dB for the control, and the control itself scored 9.55 dB on the two-speaker test set against 9.49 dB for the proposed model, i.e. no real gain.

4.6 Limitations

- All results are on clean, read, 8 kHz English speech; noise, reverberation and conversational speech were not evaluated.
- The four-speaker test set is this project's own extension, not an official benchmark.

- The model is non-causal; streaming relies on a sliding window, which sets a lower bound of a few hundred milliseconds on latency.
- The attention study is preliminary (Section 4.2): the runs were stopped early, so only the first epochs of training are compared.
- On-device results come from one phone model and the accurate (R = 6) model with 4 threads; power and battery consumption were not measured.

CHAPTER 5
Conclusion and Future Work

5.1 Conclusion

This project built a complete target speech extraction system for edge devices: a reproducible data and training pipeline, a compact speaker-conditioned U-shaped convolutional network, an evaluation suite with eight metrics, and a deployment path through ONNX export, INT8 quantisation and sliding-window streaming.

The main finding is that one model trained with dynamic 2-4 speaker mixing handles an unknown number of talkers: 9.49, 8.13 and 7.00 dB SI-SDRi on two, three and four speakers, whereas the same design trained on two speakers only drops to 3.13 dB on four. The improvement comes from suppressing the additional talkers (higher SIRi), and it carries over to perceived quality and intelligibility (PESQ and STOI). The remaining errors are wrong-speaker extractions between similar voices, which two ECAPA-based attempts did not fix, pointing to the speaker cue as the next lever.

For deployment, the network splits naturally into a once-per-speaker embedder and a streaming extractor; the exported ONNX graphs reproduce the PyTorch outputs, and INT8 quantisation reduces the model to about a quarter of its size with a small loss in quality. Replacing the channel attention of the separator bottleneck by multi-head or differential attention did not help with the same training recipe (the runs fell steadily behind and were stopped early), although differential attention lagged less than standard attention. On a commercial smartphone (Samsung SM-S731B) the offline Android application runs the proposed model 13 times faster than real time, reproduces the PC extraction quality (9.84 vs 9.83 dB) and streams in real time with 1 s of latency, while the compact model streams with 0.4 s of latency on a single CPU core - target speech extraction on an edge device is practical today.

5.2 Future Work

- **Cross-corpus training and testing on TIMIT.** Training jointly on LibriSpeech and TIMIT (with interferers drawn from the same corpus to avoid a recording-condition shortcut) and testing on unseen TIMIT speakers would measure how well the model generalises beyond read audiobooks.

- **Revisit bottleneck attention** with a learning-rate warm-up and a lower learning rate for the attention layers (usual for Transformer blocks), or by fine-tuning the trained model with attention added, and evaluate it on the full test sets.
- **A causal model** trained for streaming (causal convolutions, cumulative normalisation) to bring the latency below 150 ms.
- **A stronger speaker cue** for similar voices, for example contrastive training of the speaker encoder on more speakers (LibriSpeech train-clean-360) or enrollment-mixture cross-attention.
- **Noise and reverberation** (e.g. the noisy LibriMix variants) and real recordings.
- **Quantisation-aware training** and hardware-specific runtimes (e.g. NPU delegates) on the edge device.

Appendix

The complete source code (about 6,000 lines of Python with 81 automated tests) is in the project repository https://github.com/im-Johaan/PJT1-TSE, together with the Android application and all result files. The appendices list the parts most relevant to this report.

Appendix 1: Bottleneck Attention Modules (src/tse/models/layers.py)

```
class MultiHeadAttention(nn.Module):
    """Pre-norm multi-head self-attention over time with a residual path.

    Tokens are the frames of ``x`` (B, C, T); queries/keys/values live in
``attn_dim``
    (< C keeps it cheap). The output projection is zero-initialised, so the
layer is the
    identity at initialisation.
    """

    def __init__(self, channels: int, attn_dim: int = 256, heads: int = 4) -
> None:
        super().__init__()
        if attn_dim % heads:
            raise ValueError(f"attn_dim {attn_dim} not divisible by heads
{heads}")
        self.heads = heads
        self.norm = nn.LayerNorm(channels)
        self.qkv = nn.Linear(channels, 3 * attn_dim)
        self.out = nn.Linear(attn_dim, channels)
        nn.init.zeros_(self.out.weight)
        nn.init.zeros_(self.out.bias)

    def _split(self, t: torch.Tensor) -> torch.Tensor:
        """(B, T, D) -> (B, heads, T, D / heads)."""
        Bn, T, D = t.shape
        return t.view(Bn, T, self.heads, D // self.heads).transpose(1, 2)

    def attend(self, h: torch.Tensor) -> torch.Tensor:
        """(B, T, C) normalised tokens -> (B, heads, T, D / heads) attention
output."""
        q, k, v = (self._split(t) for t in self.qkv(h).chunk(3, dim=-1))
        return F.scaled_dot_product_attention(q, k, v)

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        """x: (B, C, T) -> (B, C, T)."""
        h = self.norm(x.transpose(1, 2))
        o = self.attend(h)
        o = o.transpose(1, 2).reshape(h.shape[0], h.shape[1], -1)
        return x + self.out(o).transpose(1, 2)



class DifferentialAttention(MultiHeadAttention):
    """Differential attention (Ye et al., 2024, "Differential Transformer")
over time.

    Each head computes two softmax maps from split queries/keys and uses
their difference
    ``(softmax(Q1 K1^T) - lambda * softmax(Q2 K2^T)) V``, which cancels
attention mass both
    maps share (common-mode "noise"). ``lambda`` is re-parameterised as
    ``exp(lq1.lk1) - exp(lq2.lk2) + lambda_init`` with ``lambda_init`` set
by ``depth``; each
```

```
    head output is RMS-normalised and scaled by ``1 - lambda_init``.

    Uses ``heads // 2`` heads of twice the width, so parameters match
    :class:`MultiHeadAttention` with the same ``attn_dim`` and ``heads``.
    """

    def __init__(self, channels: int, attn_dim: int = 256, heads: int = 4,
depth: int = 0) -> None:
        if heads % 2:
            raise ValueError(f"differential attention needs an even number
of heads, got {heads}")
        super().__init__(channels, attn_dim, heads // 2)
        dh = attn_dim // self.heads // 2  # width of each of the two
query/key halves
        self.lambda_init = 0.8 - 0.6 * math.exp(-0.3 * depth)
        self.lq1, self.lk1, self.lq2, self.lk2 = (
            nn.Parameter(torch.randn(dh) * 0.1) for _ in range(4))
        self.head_norm = nn.RMSNorm(2 * dh, eps=1e-5,
elementwise_affine=False)

    def attend(self, h: torch.Tensor) -> torch.Tensor:
        q, k, v = (self._split(t) for t in self.qkv(h).chunk(3, dim=-1))
        q1, q2 = q.chunk(2, dim=-1)
        k1, k2 = k.chunk(2, dim=-1)
        lam = (torch.exp((self.lq1 * self.lk1).sum()) - torch.exp((self.lq2
* self.lk2).sum())
               + self.lambda_init)
        o = (F.scaled_dot_product_attention(q1, k1, v)
             - lam * F.scaled_dot_product_attention(q2, k2, v))
        return self.head_norm(o) * (1.0 - self.lambda_init)
```

Appendix 2: U-ConvBlock Separator Stage (src/tse/models/ublock.py)

```
class UConvBlock(nn.Module):
    """One separator stage: U-shaped multi-resolution conv block with a
residual path.

    ``x (B, Bc, F) -> 1x1 (Bc->H) -> Q stride-2 DS-conv downsamplings ->
fusion(e) +
    attention at F/2^Q -> upsample + skip-sum back to F -> 1x1 (H->Bc) -> +
x``.

    ``attention`` is ``se`` (squeeze-and-excitation channel attention),
``mhsa``
    (multi-head self-attention over time) or ``diff`` (differential
attention over time);
    ``depth`` (stage index) sets the differential-attention ``lambda_init``.

    The time axis is zero-padded to a multiple of ``2**Q`` internally and
cropped back,
    so any length works.
    """

    def __init__(self, B: int, H: int, Q: int, K: int, spk_dim: int, fusion:
str = "film",
                 upsample: str = "nearest", se_reduction: int = 8,
attention: str = "se",
                 attn_dim: int = 256, attn_heads: int = 4, depth: int = 0) -
> None:
        super().__init__()
        self.Q = Q
        self.proj_in = nn.Sequential(nn.Conv1d(B, H, 1), nn.PReLU(),
GlobalLayerNorm(H))
```

```
        self.down = nn.ModuleList([DSConvBlock(H, H, K, stride=2) for _ in
range(Q)])
        self.fusion = make_fusion(fusion, spk_dim, H)
        self.attn = make_attention(attention, H, se_reduction, attn_dim,
attn_heads, depth)
        self.upsample = upsample
        if upsample == "transposed":
            self.up = nn.ModuleList([nn.ConvTranspose1d(H, H, 4, stride=2,
padding=1, groups=H)
                                     for _ in range(Q)])
        self.proj_out = nn.Conv1d(H, B, 1)

    def _up(self, y: torch.Tensor, q: int, size: int) -> torch.Tensor:
        if self.upsample == "transposed":
            y = self.up[q](y)
            return y[..., :size]
        return F.interpolate(y, size=size, mode="nearest")

    def forward(self, x: torch.Tensor, e: torch.Tensor | None) ->
torch.Tensor:
        """x: (B, Bc, F); e: (B, D) or None (no conditioning in this
stage)."""
        T = x.shape[-1]
        mult = 2 ** self.Q
        pad = (mult - T % mult) % mult
        y = self.proj_in(F.pad(x, (0, pad)))
        skips = [y]
        for d in self.down:
            y = d(y)
            skips.append(y)
        if e is not None:
            y = self.fusion(y, e)
        y = self.attn(y)
        for q in reversed(range(self.Q)):
            y = self._up(y, q, skips[q].shape[-1]) + skips[q]
        return self.proj_out(y)[..., :T] + x
```

Appendix 3: BSS-eval SIR / SAR Implementation (src/tse/metrics.py)

```
def _xcorr(a: np.ndarray, b: np.ndarray, n_lags: int, nfft: int) ->
np.ndarray:
    """``r[t] = sum_n a[n] b[n + t]`` for lags ``t = 0 .. n_lags-1`` (rows
of a, b broadcast)."""
    fa, fb = np.fft.rfft(a, nfft), np.fft.rfft(b, nfft)
    return np.fft.irfft(np.conj(fa) * fb, nfft)[..., :n_lags]



def bss_eval_target(est: np.ndarray, ref: np.ndarray, interferers:
np.ndarray,
                    filter_length: int = 512, load_diag: float = 1e-10
                    ) -> tuple[float, float, float]:
    """BSS-eval SDR, SIR and SAR of ``est`` for target ``ref`` (Vincent et
al., 2006; the
    ``bss_eval_sources`` definition of mir_eval / BSS Eval v3 with 512-tap
filters).

    ``interferers`` (K, T) holds each interfering source separately. The
estimate is
    projected (least squares, time-invariant FIR filters) onto the target
alone and onto
    all references: ``s_target = P_target est``, ``e_interf = P_all est -
s_target``,
    ``e_artif = est - P_all est``; ``SDR = |s_target|^2 / |e_interf +
e_artif|^2``,
```

```
    ``SIR = |s_target|^2 / |e_interf|^2`` and ``SAR = |s_target +
e_interf|^2 / |e_artif|^2``.
    SDR depends only on the target projection, so it equals fast_bss_eval's
single-reference
    ``sdr`` (used for the SDR numbers reported before SIR/SAR were added).

    Implemented directly in numpy/scipy (Cholesky solve of the block-
Toeplitz Gram matrix):
    fast_bss_eval's numpy backend fails on numpy>=2 with several references
and its torch
    backend hits an MKL ``lu_solve`` error on some inputs.
    """
    L = filter_length
    refs = np.concatenate([ref[None], interferers]).astype(np.float64)
    x = est.astype(np.float64)
    n, T = refs.shape
    nfft = int(2 ** np.ceil(np.log2(T + L)))
    # Gram matrix G[(i,a),(j,b)] = sum_t r_i[t-a] r_j[t-b] = R_ij[a-b]
(zero-padded signals)
    G = np.empty((n * L, n * L))
    for i in range(n):
        for j in range(i, n):
            r_ij = _xcorr(refs[i], refs[j], L, nfft)  # sum r_i[t]
r_j[t+lag], lag >= 0
            r_ji = _xcorr(refs[j], refs[i], L, nfft)
            lag = np.arange(L)[:, None] - np.arange(L)[None, :]  # a - b
            blk = np.where(lag >= 0, r_ij[np.abs(lag)], r_ji[np.abs(lag)])
            G[i * L:(i + 1) * L, j * L:(j + 1) * L] = blk
            G[j * L:(j + 1) * L, i * L:(i + 1) * L] = blk.T
    # D[(i,a)] = sum_t r_i[t-a] x[t]
    D = _xcorr(refs, x[None], L, nfft).reshape(-1)
    reg = load_diag * np.trace(G) / len(G) + 1e-12
    G[np.diag_indices_from(G)] += reg

    def project(k: int) -> np.ndarray:
        c = scipy.linalg.cho_solve(scipy.linalg.cho_factor(G[:k * L, :k *
L]), D[:k * L])
        return sum(scipy.signal.fftconvolve(refs[i], c[i * L:(i + 1) *
L])[:T] for i in range(k))

    s_target = project(1)
    p_all = project(n)
    e_interf = p_all - s_target
    e_artif = x - p_all
    eps = 1e-20
    sir = 10 * np.log10((s_target @ s_target + eps) / (e_interf @ e_interf +
eps))
    sar = 10 * np.log10((p_all @ p_all + eps) / (e_artif @ e_artif + eps))
    e_total = x - s_target
    sdr = 10 * np.log10((s_target @ s_target + eps) / (e_total @ e_total +
eps))
    return float(sdr), float(sir), float(sar)
```

Appendix 4: On-Device Inference Engine of the Android Application (TseEngine.java)

```
    public TseEngine(byte[] embedderModel, byte[] extractorModel, int
threads) throws OrtException {
        long t0 = System.nanoTime();
        this.threads = threads;
        OrtSession.SessionOptions so = new OrtSession.SessionOptions();
        so.setIntraOpNumThreads(threads);
        so.setInterOpNumThreads(1);
        so.setOptimizationLevel(OrtSession.SessionOptions.OptLevel.ALL_OPT);
```

```
        embedder = env.createSession(embedderModel, so);
        extractor = env.createSession(extractorModel, so);
        loadMillis = (System.nanoTime() - t0) / 1_000_000;
    }

    /** Speaker embedding (256 floats) of an enrollment recording. */
    public float[] embed(float[] enrollment) throws OrtException {
        try (OnnxTensor x = tensor(enrollment);
             OrtSession.Result r =
embedder.run(Collections.singletonMap("enrollment", x))) {
            return ((float[][]) r.get(0).getValue())[0];
        }
    }

    /** Target estimate for a mixture (same length; arbitrary output level,
see {@link #matchLevel}). */
    public float[] extract(float[] mixture, float[] embedding) throws
OrtException {
        try (OnnxTensor x = tensor(mixture); OnnxTensor e =
tensor(embedding)) {
            java.util.HashMap<String, OnnxTensor> in = new
java.util.HashMap<>();  // Map.of needs Android 11+
            in.put("mixture", x);
            in.put("embedding", e);
            try (OrtSession.Result r = extractor.run(in)) {
                return ((float[][]) r.get(0).getValue())[0];
            }
        }
    }
```

Appendix 5: Reproducing the Results

```
# environment
pip install torch torchaudio --index-url
https://download.pytorch.org/whl/cu128
pip install -r requirements.txt pesq onnx onnxruntime
# training (proposed model; attention variants)
python scripts/train.py --config configs/multispk_deep.yaml
powershell -File experiments/attention/run.ps1 diff
# evaluation on 2/3/4 speakers with all metrics
powershell -File experiments/report_eval/run.ps1
# edge: export, quantise, accuracy and CPU latency
python scripts/edge_benchmark.py --models
r6=saved_models/multispk_r6_ep79/best.pt r3=runs/finetune_multispk/best.pt
python scripts/stream_eval.py --model r6_multispk --sets 2spk 3spk 4spk --n
200
# Android application (offline, models inside the APK)
python phone_demo/prepare_phone_assets.py; python
android_app/prepare_assets.py
bash android_app/build_apk.sh          # -> android_app/build/TSE-Edge.apk
# figures and this report
python report/diagrams.py; python report/figures.py; python
report/build_report.py
```

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